KASHYAP PANDEY — 3-MONTH PREP PLAN

Principal / Staff SDE | MAANG · Amex · eBay · Visa · Uber | March 1 – May 31, 2026

SESSION FORMAT

Weekday — 2 hrs	Saturday — 5 hrs	Sunday — 5 hrs

How to start every day: tell me "Today is [DATE] — let's begin" and I'll load your session automatically.

MONTH 1: MARCH 2026 — Build the Muscle: DSA Foundations + System Design Intro

WEEK 1 (Mar 1–7) | Arrays · Sliding Window · Binary Search · Linked Lists

Sun, Mar 1 | *WEEKEND — 5 hrs • ORIENTATION DAY*
Kickoff + Arrays Foundations + System Design Intro

Hour 1–2: Orientation — review 3-month plan, set up LeetCode, solve Two Sum + Best Time to Buy/Sell
Hour 3–4: Arrays Deep Dive — Prefix Sum, Kadane's Algorithm, Product of Array Except Self
Hour 5: System Design Preview — distributed systems vocab, CAP theorem, consistency vs availability

Start Command:
"Today is Mar 1 — Orientation Day, let's begin"

Mon, Mar 2 | *WEEKDAY — 2 hrs*
Sliding Window Pattern

0:00–0:10 Concept: fixed vs variable window
0:10–0:50 Problem 1: Longest Substring Without Repeating Characters
0:50–1:30 Problem 2: Minimum Window Substring
1:30–2:00 Review + pattern notes

Start Command:
"Today is Mar 2 — Sliding Window, let's begin"

Tue, Mar 3 | *WEEKDAY — 2 hrs*
Two Pointer Technique

0:00–0:10 Concept: opposite ends vs slow-fast pointers
0:10–0:50 Problem 1: 3Sum
0:50–1:30 Problem 2: Container With Most Water
1:30–2:00 Review

Start Command:
"Today is Mar 3 — Two Pointers, let's begin"

Wed, Mar 4 | *WEEKDAY — 2 hrs*
Binary Search + Variants

0:00–0:10 Concept: classic + rotated + answer-space binary search
0:10–0:50 Problem 1: Search in Rotated Sorted Array
0:50–1:30 Problem 2: Find Minimum in Rotated Sorted Array

Start Command:
"Today is Mar 4 — Binary Search, let's begin"

1:30–2:00 Review

Thu, Mar 5 | *WEEKDAY — 2 hrs*
Linked List Core Operations

0:00–0:10 Concept: dummy nodes, runner technique, in-place reversal
0:10–0:50 Problem 1: Reverse Linked List (iterative + recursive)
0:50–1:30 Problem 2: Merge Two Sorted Lists
1:30–2:00 Review

Start Command:
"Today is Mar 5 — Linked Lists, let's begin"

Fri, Mar 6 | *WEEKDAY — 2 hrs*
Mixed Practice: Arrays + Sliding Window + Two Pointer

0:00–0:55 Problem 1: Subarray Sum Equals K (timed 25 min)
0:55–1:50 Problem 2: Trapping Rain Water (timed 30 min)
1:50–2:00 Week 1 review — what's sticky, what needs work

Start Command:
"Today is Mar 6 — Mixed Practice, let's begin"

Sat, Mar 7 | *WEEKEND — 5 hrs • SYSTEM DESIGN DEEP DIVE*
Design a Rate Limiter

Hour 1: Requirements gathering (functional + non-functional)
Hour 2: Algorithms — Token Bucket, Leaky Bucket, Fixed Window, Sliding Window Counter
Hour 3: Distributed rate limiting — Redis, sticky sessions, race conditions
Hour 4: HLD diagram + API design + DB schema
Hour 5: Trade-offs mock Q&A — 'Why not just use API Gateway?'

Start Command:
"Today is Mar 7 — System Design: Rate Limiter, let's begin"

WEEK 2 (Mar 8–14) | Stacks · Queues · Linked List Advanced · Strings

Sun, Mar 8 | *WEEKEND — 5 hrs • MOCK #1 + LINKED LIST DEEP*
Mock Coding Interview + LRU Cache + Rate Limiter Review

Hour 1–2: Mock coding interview — 2 timed problems (35 min each), full debrief
Hour 3–4: Linked List Advanced — Floyd's cycle detection, Reorder List, LRU Cache
Hour 5: Revisit Rate Limiter design, close any gaps

Start Command:
"Today is Mar 8 — Mock #1 + Linked List Deep, let's begin"

Mon, Mar 9 | *WEEKDAY — 2 hrs*
Stack Patterns — Monotonic Stack

0:00–0:10 Concept: when to use monotonic stack (next greater, spans, histogram)
0:10–0:50 Problem 1: Daily Temperatures
0:50–1:30 Problem 2: Largest Rectangle in Histogram
1:30–2:00 Review

Start Command:
"Today is Mar 9 — Stacks, let's begin"

Tue, Mar 10 | *WEEKDAY — 2 hrs*
Queues + Deque Patterns

0:00–0:10 Concept: monotonic deque, BFS queue
0:10–0:50 Problem 1: Sliding Window Maximum
0:50–1:30 Problem 2: Number of Recent Calls
1:30–2:00 Review

Start Command:
"Today is Mar 10 — Queues & Deque, let's begin"

Wed, Mar 11 | *WEEKDAY — 2 hrs*
Linked List Advanced

0:00–0:10 Concept: multi-pointer, in-place manipulation
0:10–0:50 Problem 1: Copy List with Random Pointer
0:50–1:30 Problem 2: Find the Duplicate Number (Floyd's + array)
1:30–2:00 Review

Start Command:
"Today is Mar 11 — Linked List Advanced, let's begin"

Thu, Mar 12 | *WEEKDAY — 2 hrs*
String Manipulation + Hashing

0:00–0:10 Concept: anagram detection, rolling hash
0:10–0:50 Problem 1: Group Anagrams
0:50–1:30 Problem 2: Longest Palindromic Substring
1:30–2:00 Review

Start Command:
"Today is Mar 12 — String Patterns, let's begin"

Fri, Mar 13 | *WEEKDAY — 2 hrs*
Mixed Practice: Stack + Queue + String

0:00–0:55 Problem 1: Valid Parentheses → extend to Minimum Remove to Make Valid
0:55–1:50 Problem 2: Decode String (stack-based)
1:50–2:00 Review

Start Command:
"Today is Mar 13 — Mixed Practice, let's begin"

Sat, Mar 14 | *WEEKEND — 5 hrs • SYSTEM DESIGN DEEP DIVE*
Design URL Shortener (TinyURL)

Hour 1: Scale estimation — 100M URLs/day, read-heavy 100:1 ratio
Hour 2: Hashing strategies — MD5, Base62, collision handling
Hour 3: DB design — SQL vs NoSQL, Redis caching layer
Hour 4: Scalability — CDN, geo-distribution, analytics tracking
Hour 5: Mock Q&A — 'How do you handle hash collisions at scale?'

Start Command:
"Today is Mar 14 — System Design: URL Shortener, let's begin"

WEEK 3 (Mar 15–21) | Binary Trees · BST · Tree Construction

Sun, Mar 15 | *WEEKEND — 5 hrs • MOCK #2 + TREES INTRO*
Mock Coding + Binary Tree Foundations

Hour 1–2: Mock coding — 1 sliding window + 1 stack problem (timed)
Hour 3–5: Binary Tree foundations — DFS/BFS traversals, Maximum Depth, Symmetric Tree, Path Sum

Start Command:
"Today is Mar 15 — Mock #2 + Trees Intro, let's begin"

Mon, Mar 16 | *WEEKDAY — 2 hrs*
Binary Tree DFS Patterns

0:00–0:10 Concept: top-down vs bottom-up recursion
0:10–0:50 Problem 1: Lowest Common Ancestor of a Binary Tree
0:50–1:30 Problem 2: Binary Tree Maximum Path Sum
1:30–2:00 Review

Start Command:
"Today is Mar 16 — Binary Tree DFS, let's begin"

Tue, Mar 17 | *WEEKDAY — 2 hrs*
Binary Tree BFS — Level Order

0:00–0:10 Concept: queue-based BFS, zigzag, right side view
0:10–0:50 Problem 1: Binary Tree Right Side View
0:50–1:30 Problem 2: Populating Next Right Pointers in Each Node
1:30–2:00 Review

Start Command:
"Today is Mar 17 — Binary Tree BFS, let's begin"

Wed, Mar 18 | *WEEKDAY — 2 hrs*
Binary Search Tree Operations

0:00–0:10 Concept: BST invariant, in-order traversal = sorted
0:10–0:50 Problem 1: Validate Binary Search Tree
0:50–1:30 Problem 2: Kth Smallest Element in a BST
1:30–2:00 Review

Start Command:
"Today is Mar 18 — BST, let's begin"

Thu, Mar 19 | *WEEKDAY — 2 hrs*
Construct Trees from Traversals

0:00–0:10 Concept: using preorder + inorder to rebuild a tree
0:10–0:50 Problem 1: Construct Binary Tree from Preorder and Inorder Traversal
0:50–1:30 Problem 2: Serialize and Deserialize Binary Tree
1:30–2:00 Review

Start Command:
"Today is Mar 19 — Tree Construction, let's begin"

Fri, Mar 20 | *WEEKDAY — 2 hrs*
Mixed Tree Practice

0:00–0:55 Problem 1: Diameter of Binary Tree
0:55–1:50 Problem 2: Count Good Nodes in Binary Tree
1:50–2:00 Review

Start Command:
"Today is Mar 20 — Mixed Tree Practice, let's begin"

Sat, Mar 21 | *WEEKEND — 5 hrs • SYSTEM DESIGN DEEP DIVE*
Design a Push Notification System

Hour 1: Requirements — multi-channel (SMS, email, push), 10M notifications/day
Hour 2: Architecture — message queue, channel workers, adapters
Hour 3: Reliability — retries, deduplication, priority queues, dead-letter queues
Hour 4: DB design + API contracts
Hour 5: Mock Q&A + trade-offs discussion

Start Command:
"Today is Mar 21 — System Design: Notification System, let's begin"

WEEK 4 (Mar 22–28) | Graphs · Union-Find · DP Introduction

Sun, Mar 22 | *WEEKEND — 5 hrs • MOCK #3 + GRAPH INTRO*
Mock Coding + Graph Foundations

Hour 1–2: Mock coding — 1 BST + 1 tree BFS problem (timed + scored)
Hour 3–5: Graph foundations — adjacency list, DFS/BFS, Number of Islands, Clone Graph, Course Schedule

Start Command:
"Today is Mar 22 — Mock #3 + Graph Intro, let's begin"

Mon, Mar 23 | *WEEKDAY — 2 hrs*
Graph BFS + Shortest Path

0:00–0:10 Concept: BFS guarantees shortest path in unweighted graphs
0:10–0:50 Problem 1: Rotting Oranges
0:50–1:30 Problem 2: Word Ladder
1:30–2:00 Review

Start Command:
"Today is Mar 23 — Graph BFS, let's begin"

Tue, Mar 24 | *WEEKDAY — 2 hrs*
Topological Sort + Cycle Detection

0:00–0:10 Concept: Kahn's algorithm (BFS) + DFS approach
0:10–0:50 Problem 1: Course Schedule II
0:50–1:30 Problem 2: Alien Dictionary
1:30–2:00 Review

Start Command:
"Today is Mar 24 — Topological Sort, let's begin"

Wed, Mar 25 | *WEEKDAY — 2 hrs*
Union-Find (Disjoint Set Union)

0:00–0:10 Concept: path compression + union by rank
0:10–0:50 Problem 1: Number of Connected Components in Undirected Graph
0:50–1:30 Problem 2: Accounts Merge
1:30–2:00 Review

Start Command:
"Today is Mar 25 — Union-Find, let's begin"

Thu, Mar 26 | *WEEKDAY — 2 hrs*
DP Fundamentals — Intro

0:00–0:10 Concept: overlapping subproblems, optimal substructure, memoization vs tabulation
0:10–0:50 Problem 1: Climbing Stairs → House Robber
0:50–1:30 Problem 2: Coin Change
1:30–2:00 Review + DP template

Start Command:
"Today is Mar 26 — DP Introduction, let's begin"

Fri, Mar 27 | *WEEKDAY — 2 hrs*
DP — 1D Problems

0:00–0:55 Problem 1: Longest Increasing Subsequence
0:55–1:50 Problem 2: Word Break
1:50–2:00 Review

Start Command:
"Today is Mar 27 — DP 1D, let's begin"

Sat, Mar 28 | *WEEKEND — 5 hrs • SYSTEM DESIGN DEEP DIVE*

Design a Distributed Key-Value Store (like Redis)

Hour 1: Requirements + consistency model choices (strong vs eventual)
Hour 2: Partitioning — consistent hashing, virtual nodes
Hour 3: Replication — leader-follower, quorum reads/writes
Hour 4: Fault tolerance — gossip protocol, vector clocks, anti-entropy
Hour 5: Mock Q&A — 'How is this different from how Kafka stores offsets?'

Start Command:
"Today is Mar 28 — System Design: Key-Value Store, let's begin"

Sun, Mar 29 | *WEEKEND — 5 hrs • MONTH 1 REVIEW + MOCK #4*
Month 1 Consolidation + Preview of Month 2

Hour 1–2: Full mock coding (2 surprise problems from Month 1 — timed, scored)
Hour 3: Re-solve your 3 weakest problems; consolidate DSA patterns
Hour 4: System Design cold review — explain 2 designs from scratch without notes
Hour 5: Month 2 preview — what's coming in April; set your goals

Start Command:
"Today is Mar 29 — Month 1 Review + Mock #4, let's begin"

Mon, Mar 30 | *WEEKDAY — 2 hrs*
DP — 2D Grid Problems

0:00–0:10 Concept: grid DP state definition and setup
0:10–0:50 Problem 1: Unique Paths
0:50–1:30 Problem 2: Minimum Path Sum
1:30–2:00 Review

Start Command:
"Today is Mar 30 — DP 2D, let's begin"

Tue, Mar 31 | *WEEKDAY — 2 hrs*
DP — Subsequences (LCS + Edit Distance)

0:00–0:10 Concept: classic 2D DP table construction
0:10–0:50 Problem 1: Longest Common Subsequence
0:50–1:30 Problem 2: Edit Distance
1:30–2:00 Review

Start Command:
"Today is Mar 31 — DP Subsequences, let's begin"

MONTH 2: APRIL 2026 — Go Wide and Deep: Advanced DSA + System Design at Scale

WEEK 5 (Apr 1–7) | Heaps · DP Knapsack · Tries · Backtracking

Wed, Apr 1 | *WEEKDAY — 2 hrs*
Heap / Priority Queue Patterns

0:00–0:10 Concept: min-heap vs max-heap, heapify in O(n)
0:10–0:50 Problem 1: Kth Largest Element in an Array
0:50–1:30 Problem 2: Merge K Sorted Lists
1:30–2:00 Review

Start Command:
"Today is Apr 1 — Heaps, let's begin"

Thu, Apr 2 | *WEEKDAY — 2 hrs*
Two-Heap Pattern + Stream Problems

0:00–0:10 Concept: two heaps for median, sliding window with heap
0:10–0:50 Problem 1: Find Median from Data Stream
0:50–1:30 Problem 2: Task Scheduler
1:30–2:00 Review

Start Command:
"Today is Apr 2 — Heaps Advanced, let's begin"

Fri, Apr 3 | *WEEKDAY — 2 hrs*
DP — 0/1 Knapsack + Variants

0:00–0:10 Concept: include/exclude decision tree
0:10–0:50 Problem 1: Partition Equal Subset Sum
0:50–1:30 Problem 2: Target Sum
1:30–2:00 Review

Start Command:
"Today is Apr 3 — DP Knapsack, let's begin"

Sat, Apr 4 | *WEEKEND — 5 hrs • SYSTEM DESIGN DEEP DIVE*
Design Kafka / Distributed Message Queue from Scratch

Hour 1: Requirements — throughput, ordering, durability, at-least-once delivery
Hour 2: Core internals — partitions, offsets, log segments, compaction
Hour 3: Replication — ISR, leader election, Raft/ZooKeeper
Hour 4: Consumer groups, rebalancing, exactly-once semantics
Hour 5: Mock Q&A — 'How would you add multi-tenancy to Kafka?'

Start Command:
"Today is Apr 4 — System Design: Kafka Deep Dive, let's begin"

Sun, Apr 5 | *WEEKEND — 5 hrs • MOCK #5 + BEHAVIORAL ROUND 1*
Coding Mock + Amazon LP Behavioral

Hour 1–2: Mock coding — 1 heap + 1 DP medium (timed)
Hour 3–4: Behavioral mock — 5 Amazon LP questions, STAR-I format, I play interviewer
Hour 5: Debrief + polish your top 3 behavioral stories

Start Command:
"Today is Apr 5 — Mock #5 + Behavioral Round 1, let's begin"

Mon, Apr 6 | *WEEKDAY — 2 hrs*
Trie Data Structure

0:00–0:10 Concept: trie node structure, insert, search, startsWith
0:10–0:50 Problem 1: Implement Trie (Prefix Tree)
0:50–1:30 Problem 2: Word Search II
1:30–2:00 Review

Start Command:
"Today is Apr 6 — Tries, let's begin"

Tue, Apr 7 | *WEEKDAY — 2 hrs*
Backtracking Fundamentals

0:00–0:10 Concept: decision tree, pruning strategies
0:10–0:50 Problem 1: Combination Sum
0:50–1:30 Problem 2: Permutations
1:30–2:00 Review

Start Command:
"Today is Apr 7 — Backtracking, let's begin"

WEEK 6 (Apr 8–14) | Backtracking Hard · Java Concurrency · Intervals

Wed, Apr 8 | *WEEKDAY — 2 hrs*
Backtracking — Hard Problems

0:00–0:10 Concept: state pruning, visited sets
0:10–0:50 Problem 1: N-Queens
0:50–1:30 Problem 2: Sudoku Solver
1:30–2:00 Review

Start Command:
"Today is Apr 8 — Backtracking Advanced, let's begin"

Thu, Apr 9 | *WEEKDAY — 2 hrs*
Java Concurrency for Interviews

0:00–0:10 Concept: thread safety, happens-before, volatile, synchronized
0:10–0:50 Design: Implement Thread-Safe LRU Cache
0:50–1:30 Design: Implement Blocking Queue from scratch
1:30–2:00 Review + Java memory model essentials

Start Command:
"Today is Apr 9 — Java Concurrency, let's begin"

Fri, Apr 10 | *WEEKDAY — 2 hrs*
Mixed Hard Practice

0:00–0:55 Problem 1: Longest Consecutive Sequence
0:55–1:50 Problem 2: Pacific Atlantic Water Flow
1:50–2:00 Review

Start Command:
"Today is Apr 10 — Mixed Hard Practice, let's begin"

Sat, Apr 11 | *WEEKEND — 5 hrs • SYSTEM DESIGN DEEP DIVE*
Design Uber / Ride Sharing Platform

Hour 1: Requirements + scale — 10M rides/day, real-time driver location
Hour 2: Geo-indexing — QuadTree, H3 hex grid, PostGIS
Hour 3: Ride matching — supply-demand proximity + ETA calculation
Hour 4: Surge pricing — demand zones, price multiplier logic, fairness
Hour 5: Real-time — WebSockets for live location streaming

Start Command:
"Today is Apr 11 — System Design: Uber, let's begin"

Sun, Apr 12 | *WEEKEND — 5 hrs • MOCK #6 + SYSTEM DESIGN MOCK*
Coding Mock + Design WhatsApp

Hour 1–2: Mock coding — 1 backtracking + 1 graph (timed, no hints)
Hour 3–4: System Design Mock — 'Design WhatsApp' (you present end-to-end, I ask follow-ups)
Hour 5: 2 behavioral questions + re-solve your weakest problem this week

Start Command:
"Today is Apr 12 — Mock #6 + System Design Mock, let's begin"

Mon, Apr 13 | *WEEKDAY — 2 hrs*
Interval Merge + Sweep

0:00–0:10 Concept: sort by start, overlap detection
0:10–0:50 Problem 1: Merge Intervals
0:50–1:30 Problem 2: Meeting Rooms II (min rooms needed)
1:30–2:00 Review

Start Command:
"Today is Apr 13 — Intervals, let's begin"

Tue, Apr 14 | *WEEKDAY — 2 hrs*
Intervals Advanced + Scheduling

0:00–0:10 Concept: binary search + in-place insertion
0:10–0:50 Problem 1: Insert Interval
0:50–1:30 Problem 2: Non-overlapping Intervals
1:30–2:00 Review

Start Command:
"Today is Apr 14 — Intervals Advanced, let's begin"

WEEK 7 (Apr 15–21) | Dijkstra · Bit Manipulation · DP Trees · Behavioral

Wed, Apr 15 | *WEEKDAY — 2 hrs*
Dijkstra + Weighted Graphs

0:00–0:10 Concept: Dijkstra with min-heap, Bellman-Ford for negative weights
0:10–0:50 Problem 1: Network Delay Time
0:50–1:30 Problem 2: Cheapest Flights Within K Stops
1:30–2:00 Review

Start Command:
"Today is Apr 15 — Dijkstra, let's begin"

Thu, Apr 16 | *WEEKDAY — 2 hrs*
Bit Manipulation Tricks

0:00–0:10 Concept: XOR, masks, bit shifts
0:10–0:50 Problem 1: Single Number + Single Number II
0:50–1:30 Problem 2: Number of 1 Bits + Reverse Bits
1:30–2:00 Review

Start Command:
"Today is Apr 16 — Bit Manipulation, let's begin"

Fri, Apr 17 | *WEEKDAY — 2 hrs*
Mixed Hard Practice

0:00–0:55 Problem 1: Jump Game II
0:55–1:50 Problem 2: Minimum Cost to Connect All Points (Prim's/Kruskal's)
1:50–2:00 Review

Start Command:
"Today is Apr 17 — Mixed Hard Practice, let's begin"

Sat, Apr 18 | *WEEKEND — 5 hrs • SYSTEM DESIGN DEEP DIVE*
Design a Payment Processing System (Visa / Amex Focus)

Hour 1: Requirements — authorization, settlement, global scale, regulatory compliance
Hour 2: Idempotency keys — exactly-once payments, retry safety
Hour 3: 2-Phase Commit vs Saga pattern for distributed transactions
Hour 4: Fraud detection — rule engine + real-time ML scoring layer
Hour 5: Mock Q&A — 'How do you ensure a payment is never charged twice?'

Start Command:
"Today is Apr 18 — System Design: Payments, let's begin"

Sun, Apr 19 | *WEEKEND — 5 hrs • MOCK #7 + BEHAVIORAL ROUND 2*
Hard Coding Mock + Google/Meta Behavioral

Hour 1–2: Hard coding mock — 1 hard DP + 1 graph (timed)
Hour 3–4: Behavioral — 6 Google/Meta style questions (influence, vision, failure/learning)

Start Command:
"Today is Apr 19 — Mock #7 + Behavioral Round 2, let's begin"

Hour 5: Debrief + finalize your top 10 STAR-I story bank

Mon, Apr 20 | *WEEKDAY — 2 hrs*
DP on Trees

0:00–0:10 Concept: tree DP — root choices affecting children
0:10–0:50 Problem 1: House Robber III (tree DP)
0:50–1:30 Problem 2: Binary Tree Cameras
1:30–2:00 Review

Start Command:
"Today is Apr 20 — DP on Trees, let's begin"

Tue, Apr 21 | *WEEKDAY — 2 hrs*
DP — Palindromes

0:00–0:10 Concept: expanding center vs tabulation
0:10–0:50 Problem 1: Palindromic Substrings
0:50–1:30 Problem 2: Longest Palindromic Subsequence
1:30–2:00 Review

Start Command:
"Today is Apr 21 — DP Palindromes, let's begin"

WEEK 8 (Apr 22–30) | DP State Machine · Design Patterns · E-Commerce · Full Mock

Wed, Apr 22 | *WEEKDAY — 2 hrs*
DP — State Machine (Stock Problems)

0:00–0:10 Concept: states with transitions — hold/cooldown/buy/sell
0:10–0:50 Problem 1: Best Time to Buy and Sell Stock with Cooldown
0:50–1:30 Problem 2: Best Time to Buy and Sell Stock IV
1:30–2:00 Review

Start Command:
"Today is Apr 22 — DP State Machine, let's begin"

Thu, Apr 23 | *WEEKDAY — 2 hrs*
Java Design Patterns for Principal SDE

0:00–0:10 Concept: when interviewers ask OOP/design at L6+ level
0:10–0:50 Code: Strategy Pattern + Observer Pattern with Java
0:50–1:30 Code: Decorator + Factory + Thread-Safe Singleton
1:30–2:00 Review — when to use each pattern

Start Command:
"Today is Apr 23 — Design Patterns, let's begin"

Fri, Apr 24 | *WEEKDAY — 2 hrs*
Hard Problem Simulation

0:00–0:55 Problem 1: Regular Expression Matching
0:55–1:50 Problem 2: Burst Balloons
1:50–2:00 Review

Start Command:
"Today is Apr 24 — Mixed Hard Practice, let's begin"

Sat, Apr 25 | *WEEKEND — 5 hrs • SYSTEM DESIGN DEEP DIVE*
Design an E-Commerce Marketplace (eBay Focus)

Hour 1: Requirements — catalog, inventory, bidding/listing, orders
Hour 2: Product catalog — Elasticsearch search index, faceted filtering

Start Command:

Hour 3: Inventory management — reservations, overselling prevention
Hour 4: Order workflow — Saga pattern, payment integration
Hour 5: Mock Q&A — 'How would you design eBay's real-time bidding system?'

"Today is Apr 25 — System Design: E-Commerce, let's begin"

Sun, Apr 26 | *WEEKEND — 5 hrs • FULL MOCK LOOP #1*
Complete Interview Day Simulation

Round 1 (45 min): Coding — 2 problems
Round 2 (45 min): System Design — 'Design Twitter Feed'
Round 3 (45 min): Coding — 2 harder problems
Round 4 (45 min): Behavioral — 5 questions
Debrief (60 min): Score each round, identify critical gaps

Start Command:
"Today is Apr 26 — Full Mock Loop #1, let's begin"

Mon, Apr 27 | *WEEKDAY — 2 hrs*
Month 2 Review + Weak Areas

0:00–1:00 Re-solve 2 problems you got wrong this month
1:00–2:00 Re-explain 1 system design cold (Uber or Payments, no notes)

Start Command:
"Today is Apr 27 — Month 2 Review, let's begin"

Tue, Apr 28 | *WEEKDAY — 2 hrs*
Hard Problem Blitz

0:00–0:55 Problem 1: Median of Two Sorted Arrays
0:55–1:50 Problem 2: Trapping Rain Water II (3D)
1:50–2:00 Review

Start Command:
"Today is Apr 28 — Hard Problem Blitz, let's begin"

Wed, Apr 29 | *WEEKDAY — 2 hrs*
Spring AI — Introduction

0:00–0:15 Concept: What is Spring AI? ChatClient, models, providers
0:15–0:55 Code: Build basic chat app with Spring AI + OpenAI
0:55–1:40 Code: Prompt templates + system messages
1:40–2:00 Review

Start Command:
"Today is Apr 29 — Spring AI Intro, let's begin"

Thu, Apr 30 | *WEEKDAY — 2 hrs*
Spring AI — Embeddings + Vector Stores

0:00–0:15 Concept: what are embeddings? vector similarity, cosine distance
0:15–0:55 Code: EmbeddingModel + in-memory VectorStore
0:55–1:40 Code: Store and retrieve document chunks
1:40–2:00 Review

Start Command:
"Today is Apr 30 — Spring AI Embeddings, let's begin"

MONTH 3: MAY 2026 — AI Skills + Company-Specific Prep + Perform

WEEK 10 (May 1–10) | Spring AI Deep Dive + AI Project Build

Fri, May 1 | *WEEKDAY — 2 hrs*
Spring AI — Full RAG Pipeline

0:00–0:15 Concept: DocumentReader → TextSplitter → Embed → Store → Retrieve
0:15–0:55 Code: Load PDF, chunk it, embed it, store in PGVector
0:55–1:40 Code: QuestionAnswerAdvisor — query your document
1:40–2:00 Review

Start Command:
"Today is May 1 — Spring AI RAG Pipeline, let's begin"

Sat, May 2 | *WEEKEND — 5 hrs • AI PROJECT BUILD*
Build: AI Document Classifier (ICMP-style — tie to your WF experience)

Hour 1: Architecture design for intelligent document classification
Hour 2: Code — Spring AI ChatClient + document ingestion pipeline
Hour 3: Code — classification prompt + structured output (BeanOutputConverter)
Hour 4: Code — RAG layer — retrieve similar docs + classify with context
Hour 5: Demo + 'design walkthrough' — practice explaining this in interviews

Start Command:
"Today is May 2 — AI Project Build, let's begin"

Sun, May 3 | *WEEKEND — 5 hrs • MOCK #9 + AI SYSTEM DESIGN*
Hard Coding Mock + AI Copilot System Design

Hour 1–2: Hard coding mock — 2 problems (no hints)
Hour 3–4: AI System Design — 'Design an AI Copilot for internal banking queries' (RAG, vector DB, caching, hallucination handling)
Hour 5: Debrief + story refinement

Start Command:
"Today is May 3 — Mock #9 + AI System Design, let's begin"

Mon, May 4 | *WEEKDAY — 2 hrs*
Spring AI — Tool/Function Calling

0:00–0:15 Concept: @Tool annotation, ToolCallback, agentic patterns
0:15–0:55 Code: Build a tool-calling agent (stock info fetcher)
0:55–1:40 Code: Multi-step tool invocation chain
1:40–2:00 Review

Start Command:
"Today is May 4 — Spring AI Tool Calling, let's begin"

Tue, May 5 | *WEEKDAY — 2 hrs*
Spring AI — Production Patterns

0:00–0:15 Concept: caching (prompt + response), retry, fallback patterns
0:15–0:55 Code: Implement response caching layer
0:55–1:40 Code: Observability with Micrometer — trace AI calls
1:40–2:00 Review

Start Command:
"Today is May 5 — Spring AI Production, let's begin"

Wed, May 6 | *WEEKDAY — 2 hrs*
Amazon Leadership Principles Deep Dive

0:00–1:30 Work through 8 LPs with your STAR-I stories: Ownership, Dive Deep, Invent & Simplify, Deliver Results, Bias for Action, Customer Obsession, Hire & Develop, Earn Trust
1:30–2:00 Refine 3 stories needing most polish

Start Command:
"Today is May 6 — Amazon LP Prep, let's begin"

Thu, May 7 | *WEEKDAY — 2 hrs*
Google — Staff-Level System Design Thinking

0:00–0:30 What Google looks for: scale reasoning, data modeling, back-of-envelope
0:30–1:30 Practice: 'Design Google Drive' — 1-hour compressed design
1:30–2:00 Review + gaps

Start Command:
"Today is May 7 — Google System Design, let's begin"

Fri, May 8 | *WEEKDAY — 2 hrs*
Hard Coding — MAANG Style

0:00–0:55 Problem 1: Find K Pairs with Smallest Sums
0:55–1:50 Problem 2: Minimum Window Subsequence
1:50–2:00 Review

Start Command:
"Today is May 8 — Hard Coding: MAANG, let's begin"

Sat, May 9 | *WEEKEND — 5 hrs • SYSTEM DESIGN DEEP DIVE*
Design a Real-Time Fraud Detection System (Visa / Amex)

Hour 1: Requirements — <100ms latency, accuracy, explainability, regulatory
Hour 2: Feature engineering — velocity checks, behavioral baseline, device fingerprint
Hour 3: Streaming pipeline — Kafka → Flink → ML Model serving
Hour 4: Feature store — online (Redis) vs offline (data warehouse)
Hour 5: Mock Q&A — 'How do you reduce false positives without missing fraud?'

Start Command:
"Today is May 9 — System Design: Fraud Detection, let's begin"

Sun, May 10 | *WEEKEND — 5 hrs • FULL MOCK LOOP #2*
Amazon / Google Level Simulation

Round 1 (45 min): Coding — 2 hard problems
Round 2 (45 min): System Design — 'Design Amazon Order Management'
Round 3 (45 min): Behavioral — Amazon LP (7 questions)
Round 4 (30 min): Coding — 1 hard problem
Debrief (60 min): Score all rounds, identify final gaps

Start Command:
"Today is May 10 — Full Mock Loop #2, let's begin"

WEEK 11 (May 11–17) | Company-Specific Prep + System Design Marathon + Full Mock

Mon, May 11 | *WEEKDAY — 2 hrs*
Uber — Technical Deep Dive

0:00–0:30 What Uber looks for: real-time systems, geo-spatial, scalability
0:30–1:30 Practice: 'Design Uber Eats delivery assignment' (compressed 1-hr design)
1:30–2:00 Review + gaps

Start Command:
"Today is May 11 — Uber Prep, let's begin"

Tue, May 12 | *WEEKDAY — 2 hrs*
eBay — Technical Deep Dive

0:00–0:30 What eBay looks for: marketplace, search relevance, pricing systems
0:30–1:30 Practice: 'Design eBay's real-time auction system' (compressed)
1:30–2:00 Review + gaps

Start Command:
"Today is May 12 — eBay Prep, let's begin"

Wed, May 13 | *WEEKDAY — 2 hrs*
Visa / Amex — Technical Deep Dive

0:00–0:30 What Visa/Amex look for: payments reliability, compliance, zero downtime
0:30–1:30 Practice: 'Design a credit card authorization system' (compressed)
1:30–2:00 Review + gaps

Start Command:
"Today is May 13 — Visa/Amex Prep, let's begin"

Thu, May 14 | *WEEKDAY — 2 hrs*
Resume Walkthrough + Interview Intro Practice

0:00–0:30 Practice: 'Walk me through your resume' (2 min timed, 3 versions)
0:30–1:00 Practice: 'Why do you want to move to Principal?' (polish 3 versions)
1:00–1:30 Practice: 'What's your most complex technical contribution?'
1:30–2:00 Refinement

Start Command:
"Today is May 14 — Resume Walkthrough, let's begin"

Fri, May 15 | *WEEKDAY — 2 hrs*
Final Hard Problem Blitz

0:00–0:55 Problem 1: Sliding Window + DP hybrid problem (surprise)
0:55–1:50 Problem 2: Hard Graph problem (surprise)
1:50–2:00 Review

Start Command:
"Today is May 15 — Final Hard Blitz, let's begin"

Sat, May 16 | *WEEKEND — 5 hrs • SYSTEM DESIGN MARATHON*
3 Full System Designs Back-to-Back

Design 1 (90 min): Real-time leaderboard system (Uber/eBay)
Design 2 (90 min): Distributed job scheduler (Amazon/Google)
Design 3 (60 min): Log aggregation system (any company)

Start Command:
"Today is May 16 — System Design Marathon, let's begin"

Sun, May 17 | *WEEKEND — 5 hrs • FULL MOCK LOOP #3*
Uber / eBay / Visa Level Simulation

Round 1 (45 min): Coding — 1 medium + 1 hard problem
Round 2 (45 min): System Design — 'Design a real-time payment gateway'
Round 3 (45 min): Behavioral — Uber/eBay style (impact, cross-team, scale)
Round 4 (45 min): AI System Design — 'Design an AI-powered document processor'
Debrief (60 min): Final scoring + readiness assessment

Start Command:
"Today is May 17 — Full Mock Loop #3, let's begin"

WEEK 12 (May 18–24) | Final Polish + Application Readiness

Mon, May 18 | *WEEKDAY — 2 hrs*
Weak Area Blitz

Full 2 hours: Re-solve your 4 weakest problems identified from mock loops
Focus: clean code, no bugs, verbalize every step

Start Command:
"Today is May 18 — Weak Area Blitz, let's begin"

Tue, May 19 | *WEEKDAY — 2 hrs*
Coding Speed Drill — 10 Mediums

Full 2 hours: 10 medium problems relay-style, 12 min each
Focus: speed, accuracy, clean code without overthinking

Start Command:
"Today is May 19 — Speed Drill, let's begin"

Wed, May 20 | *WEEKDAY — 2 hrs*
AI Portfolio Walkthrough

0:00–0:45 Walk through your ICMP Spring AI work (Wells Fargo context + business impact)
0:45–1:30 Walk through your AI Document Classifier project
1:30–2:00 Refine: 'How does your AI work handle hallucinations?'

Start Command:
"Today is May 20 — AI Portfolio, let's begin"

Thu, May 21 | *WEEKDAY — 2 hrs*
Final Behavioral Run — All 10 STAR-I Stories

Full 2 hours: All 10 behavioral stories — I ask, you answer, I grade
Goal: every story under 2.5 min, crisp, measurable, no rambling

Start Command:
"Today is May 21 — Behavioral Final Run, let's begin"

Fri, May 22 | *WEEKDAY — 2 hrs*
System Design Speed Round — 6 Designs

6 quick-fire designs at 20 min each: URL Shortener → Kafka → Uber → Payments → E-Commerce → Fraud Detection
Goal: can you draw the core architecture cold in 20 minutes?

Start Command:
"Today is May 22 — System Design Speed Round, let's begin"

Sat, May 23 | *WEEKEND — 5 hrs • FINAL MOCK LOOP #4*
Hardest Simulation — All Company Styles Mixed

Round 1 (45 min): Hard coding — 2 problems (surprise)
Round 2 (45 min): System Design — 'Design Slack' (surprise)
Round 3 (45 min): AI System Design — 'Add AI features to your ICMP platform'
Round 4 (45 min): Behavioral — cross-company LP mix
Debrief (90 min): Full scoring + readiness score + application checklist

Start Command:
"Today is May 23 — Final Mock Loop, let's begin"

Sun, May 24 | *WEEKEND — 5 hrs • GRADUATION DAY*
Final Review + Application Launch

Hour 1: Metrics — count total problems, review designs mastered, confirm 10 stories locked
Hour 2: Application strategy — resume per company, LinkedIn, referral approach
Hour 3–4: Final surprise mock (1 coding + 1 system design — treat it like a real interview)
Hour 5: Submit applications + set up recruiter calls

Start Command:
"Today is May 24 — Graduation Day, let's begin"

BUFFER WEEK (May 25–31) | Live Interviews + Post-Interview Debrief

These days are reserved for actual interviews and targeted re-prep based on real interview feedback. Start any of these sessions with: "Today is [DATE] — let's debrief my [Company] interview" and we'll work through what happened and sharpen your approach.

SUMMARY

Month	Weekday Sessions	Weekend Sessions	Total Hours
March	20 × 2 hrs	9 × 5 hrs	85 hrs
April	20 × 2 hrs	9 × 5 hrs	85 hrs
May	18 × 2 hrs	8 × 5 hrs	76 hrs
TOTAL	58 sessions	26 sessions	~246 hrs

START TODAY → "Today is Mar 1 — Orientation Day, let's begin"